



MD MOCHADDEC HOSSAIN

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Summary

Computer Science & Engineering student (AI/ML track) with hands-on experience in data science, NLP research, and IoT prototyping. Built end-to-end data workflows and ML prototypes (including transformer fine-tuning for harmful-content detection). Strong interest in Large Language Models and Graph Neural Networks for researcher profiling and recommendation systems. Eager to contribute as a Research Assistant to the BOREsNET project and support tasks such as data preparation, model experiments, and reproducible research.

Skills

Programming	Python · C / C++ · HTML / CSS · Kotlin
Data & Analysis	Pandas · NumPy · Matplotlib · SQL · Data cleaning / visualization
Machine Learning & NLP	scikit-learn · BERT / Transformer models · LSTM · Text preprocessing · Model evaluation
Tools	Git · Linux · Power BI · Excel · n8n (workflow automation)
Core Strengths	Research reproducibility · Analytical thinking · Data preparation · Collaboration

Professional Experience

Active Green Energy	01/2025 - 07/2025
Data Analyst Intern	
- Cleaned and formatted raw Excel data with guidance from senior accountant, ensuring accuracy - Built interactive Power BI dashboards to visualize sales and supplier data - Developed Python scripts and prediction models for business performance trends	

Education

American International University-Bangladesh (AIUB)	Dhaka
B.Sc. in Computer Science & Engineering — CGPA 3.40	08/2021 — 12/2025(expected)

Projects

Research Topic Modeling & Trend Detection
Python · BERTopic · sentence-transformers · UMAP · matplotlib
- Applied BERTopic on CS abstracts from OpenAlex to extract interpretable research topics and paper-topic mappings. - Analyzed topic evolution over years to identify emerging and declining trends. - Exported visualizations, CSV summaries, and coherence metrics for researcher profiling.
Cyberbullying Detection for Sentiment Analysis
Python · scikit-learn · NLP · Kraggle
- Built an NLP pipeline for toxic comment and sentiment detection using TF-IDF baselines (SVM, LR) and fine-tuned BERT models. - Handled preprocessing (normalization, noise removal, tokenization, class balancing) and evaluated.
Bangla-English Code-switched Slang Detection
Python · spaCy · HuggingFace · Sentence-Transformers
- Designed preprocessing for Bangla-English code-switched text (slang normalization, transliteration handling). - Trained embedding-based classifiers to detect slang and abusive variants with data augmentation for low-resource slang.

References

Md Maynul Hossain	General Manager, Active Green Energy — mdmaynul0@gmail.com
Dr. Md. Abdullah - Al - Jubair	Associate Professor, Dept. of CSE, AIUB — abdulah@aiub.edu